🚀 Complete rclone Guide

Master Multiple Google Drive Accounts & Cloud Storage Management

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🔧 Installation & Setup

Step 1: Move rclone to Permanent Location

1. Create folder: `C:\rclone`
2. Copy `rclone.exe` from your Downloads to `C:\rclone`

Step 2: Add to System PATH

1. Press **Windows key**, type **environment**
2. Click **"Edit the system environment variables"**
3. Click **"Environment Variables"** button
4. Under "System variables", find **"Path"**, click **"Edit"**

5. Click **"New"** and add: `C:\rclone`
6. Click **OK** on all windows

> ⚠️ **Important:** Close and reopen any Command Prompt windows after adding to PATH

Step 3: Verify Installation

```
# Open Command Prompt (Win + R, type cmd)
rclone version

# You should see version information
```

> ✅ **Success!** If you see version info, rclone is ready to use!

⚙️ Configure Multiple Google Drive Accounts

Initial Configuration

```
rclone config
```

Add First Google Drive (Personal Account)

1. Type `n` → Press Enter (new remote)
2. Name: `gdrive1` (or "personal", "work", whatever you like)
3. Type `drive` → Press Enter
4. Leave **Client ID** blank → Press Enter
5. Leave **Client Secret** blank → Press Enter
6. Choose `1` (Full access)
7. Leave **root_folder_id** blank → Press Enter
8. Leave **service_account_file** blank → Press Enter

9. Type `n` (No advanced config)
10. Type `y` (Use auto config) → **Browser opens**
11. **Login to your FIRST Gmail account** and authorize
12. Type `n` (Not a Shared Drive)
13. Type `y` (Yes, this is OK)
14. Type `q` (Quit config)

> 💡 **Pro Tip:** Use descriptive names like "personal", "work", "family", "client1" instead of gdrive1, gdrive2

Add Second Google Drive (Work Account)

```
rclone config
```

1. Type `n` → New remote
2. Name: `gdrive2` (or "work")
3. Follow the same steps as above
4. When browser opens, **login to your SECOND Gmail account**

Add More Accounts

Repeat the process for each Google Drive account. You can add unlimited accounts!

Example Configuration Names:

- `personal` - Your personal Gmail
- `work` - Work email
- `family` - Shared family Drive
- `client1` - Client's Drive
- `backup` - Dedicated backup Drive

View Configured Remotes

```
rclone listremotes

# Output example:
# personal:
# work:
# family:
```

Edit or Delete Remotes

```
# Edit existing remote
rclone config

# Then type 'e' and select the remote to edit

# Delete a remote
rclone config

# Then type 'd' and select the remote to delete
```

🧭 Navigation & Browsing

List All Configured Drives

```
rclone listremotes
```

Browse Root Folders

```
# List directories in root
rclone lsd personal:

# List all files and folders (with sizes)
rclone ls personal:

# List with more details (size, date)
rclone lsl personal:
```

Browse Specific Folders

```
# Navigate into a folder
rclone lsd personal:Documents

# Go deeper
rclone lsd personal:Documents/Projects

# List files in a folder
rclone ls personal:Documents/Photos

# Detailed listing
rclone lsl personal:Documents/Photos
```

Search for Files

```
# Find files by name (case-insensitive)
rclone lsf personal: --include "*report*"

# Find PDFs
rclone lsf personal: --include "*.pdf"

# Find files modified in last 7 days
rclone lsf personal: --max-age 7d
```

Get Storage Info

```
# Check space usage
rclone about personal:

# Output shows:
# Total: 15 GB
# Used: 8.5 GB
# Free: 6.5 GB
```

lsd - List Directories Only

```
rclone lsd work:
```

ls - List All Files

```
rclone ls work:
```

Shows only folders

Shows all files with sizes

📤 Transfer Operations

Understanding: Copy vs Sync vs Move

Command	What It Does	When to Use
copy	Copies files from source to destination. Source stays intact. Skips existing files.	Backup, duplicate content, one-time copy
sync	Makes destination identical to source. **Deletes files** in destination that aren't in source.	Mirror folders, keep two places identical
move	Moves files from source to destination. **Deletes from source** after successful transfer.	Transfer files, free up space in source

⚠️ **DANGER:** `sync` will DELETE files in destination that don't exist in source! Always test with `--dry-run` first!

Copy Operations

Copy Between Google Drives

```
# Copy folder from one Drive to another
rclone copy personal:Documents work:Backup

# Copy specific files only
rclone copy personal:Photos work:PhotoBackup --include "*.jpg"

# Copy with progress display
rclone copy personal:Videos work:VideoBackup -P
```

```
# Copy only new/changed files (faster)
rclone copy personal:Projects work:Projects --update
```

Copy from Local to Drive

```
# Copy local folder to Google Drive
rclone copy C:\Users\Shantu\Documents personal:Backup

# Copy specific files
rclone copy C:\Users\Shantu\Downloads\*.pdf personal:PDFs
```

Copy from Drive to Local

```
# Download folder from Drive
rclone copy personal:Photos C:\Users\Shantu\Pictures

# Download with bandwidth limit (500KB/s)
rclone copy personal:Videos C:\Users\Shantu\Videos --bwlimit 500K
```

Sync Operations

⚠️ **Always Test First:**

```
# Dry run - shows what would happen without doing it
rclone sync personal:Source work:Destination --dry-run
```

```
# Make work:Backup identical to personal:Documents
rclone sync personal:Documents work:Backup

# Sync with progress
rclone sync personal:Projects work:Projects -P

# Sync but keep deleted files in trash (safer!)
rclone sync personal:Data work:Data --drive-use-trash
```

Move Operations

```
# Move folder (deletes from source after transfer)
rclone move personal:OldFiles work:Archive

# Move specific files
rclone move personal:Downloads work:Processed --include "*.pdf"

# Move with progress
rclone move personal:Videos work:VideoArchive -P
```

Transfer Large Files

```
# For large files, use these flags:
rclone copy personal:LargeFiles work:Backup \
  --transfers 4 \
  --checkers 8 \
  --drive-chunk-size 64M \
  -P

# Explanation:
# --transfers 4: Transfer 4 files simultaneously
# --checkers 8: Check 8 files at once
# --drive-chunk-size 64M: Upload in 64MB chunks
# -P: Show progress
```

Real-World Example: Backup All Photos

```
# Backup photos from phone to two Google Drives
rclone copy C:\Users\Shantu\Phone\DCIM personal:Photos/Phone -P
rclone copy personal:Photos/Phone work:Backup/Photos -P
```

🔍 Filtering & Selection

Include/Exclude by File Type

```
# Copy only PDFs
rclone copy personal:Documents work:PDFs --include "*.pdf"

# Copy only images
rclone copy personal:Photos work:Images --include "*.{jpg,png,gif}"

# Exclude video files
rclone copy personal:Media work:Media --exclude "*.{mp4,mkv,avi}"

# Multiple includes
rclone copy personal:Files work:Files \
  --include "*.pdf" \
  --include "*.docx" \
  --include "*.xlsx"
```

Filter by Size

```
# Copy files larger than 10MB
rclone copy personal:Files work:LargeFiles --min-size 10M

# Copy files smaller than 1GB
rclone copy personal:Files work:SmallFiles --max-size 1G

# Copy files between 100KB and 50MB
rclone copy personal:Files work:MediumFiles \
  --min-size 100k \
  --max-size 50M
```

Filter by Date

```
# Copy files modified in last 7 days
rclone copy personal:Documents work:Recent --max-age 7d

# Copy files older than 30 days
rclone copy personal:Archive work:Old --min-age 30d

# Copy files from specific date range
rclone copy personal:Files work:2024Files \
  --max-age 2024-12-31 \
  --min-age 2024-01-01
```

Filter by Name Pattern

```
# Copy files starting with "Report"
rclone copy personal:Documents work:Reports --include "Report*"

# Exclude temporary files
rclone copy personal:Project work:Clean \
  --exclude "*.tmp" \
  --exclude "*.bak" \
  --exclude "*~"

# Copy files containing word "invoice"
rclone copy personal:Billing work:Invoices --include "*invoice*"
```

Exclude Folders

```
# Exclude specific folders
rclone copy personal:Project work:Project \
  --exclude "node_modules/**" \
  --exclude ".git/**" \
  --exclude "temp/**"

# Exclude hidden folders
rclone copy personal:Files work:Files --exclude ".*/"
```

Complex Filter Example

```
# Copy recent large PDFs excluding drafts
rclone copy personal:Documents work:ImportantDocs \
  --include "*.pdf" \
  --exclude "*draft*" \
  --min-size 1M \
  --max-age 90d \
  -P
```

Copies PDF files larger than 1MB, modified in last 90 days, excluding files with "draft" in name

---

⚡ Advanced Features

Speed & Performance

```
# Maximum speed (use all bandwidth)
rclone copy personal:Files work:Files --transfers 32 --checkers 32

# Limit bandwidth to 1MB/s
rclone copy personal:Files work:Files --bwlimit 1M

# Bandwidth schedule (500K during day, unlimited at night)
rclone copy personal:Files work:Files \
  --bwlimit "08:00,500k 23:00,off"

# Faster for many small files
rclone copy personal:Photos work:Photos \
  --transfers 32 \
  --fast-list
```

Progress & Logging

```
# Show progress bar
rclone copy personal:Files work:Files -P

# Show detailed transfer stats
rclone copy personal:Files work:Files --stats 5s

# Save log to file
rclone copy personal:Files work:Files --log-file=C:\rclone_log.txt

# Verbose output (debugging)
rclone copy personal:Files work:Files -vv
```

Retry & Error Handling

```
# Retry failed transfers 10 times
rclone copy personal:Files work:Files --retries 10

# Low level retries (for network issues)
rclone copy personal:Files work:Files \
  --retries 10 \
  --low-level-retries 20

# Skip files that error out
rclone copy personal:Files work:Files --ignore-errors
```

Deduplicate Files

```
# Find duplicate files
rclone dedupe personal:Files --dedupe-mode list

# Keep newest version, delete older
rclone dedupe personal:Files --dedupe-mode newest

# Keep oldest version
rclone dedupe personal:Files --dedupe-mode oldest

# Rename duplicates (safer)
rclone dedupe personal:Files --dedupe-mode rename
```

Encryption (Advanced)

```
# Create encrypted remote
rclone config

# Choose: crypt
# Base remote: personal:SecureFolder
# Set password when prompted
# Name it: encrypted

# Now use encrypted remote
rclone copy C:\Sensitive encrypted:Backup
```

💡 **Performance Tips:**

- Use `--fast-list` for folders with 1000+ files
- Increase `--transfers` for many small files
- Use `--drive-chunk-size 64M` for files >100MB
- Always use `-P` to monitor progress

💾 Mount Google Drive as Local Drive

Mount your Google Drive as a local folder (like D:, E:) to browse it in Windows Explorer!

Install WinFsp (Required for Windows)

1. Download WinFsp: https://winfsp.dev/rel/
2. Install it (default settings are fine)
3. Restart your computer

Mount Drive

```
# Mount as drive letter X:
rclone mount personal: X: --vfs-cache-mode writes

# Mount as folder
rclone mount personal: C:\MyDrive --vfs-cache-mode writes

# Mount with better caching (faster)
rclone mount personal: X: \
  --vfs-cache-mode full \
  --vfs-cache-max-size 10G \
  --vfs-cache-max-age 24h
```

Mount Multiple Drives

```
# Open multiple Command Prompt windows

# Window 1:
rclone mount personal: X: --vfs-cache-mode writes

# Window 2:
rclone mount work: Y: --vfs-cache-mode writes

# Window 3:
rclone mount family: Z: --vfs-cache-mode writes
```

⚠️ **Important:**

- Mounted drive will unmount when you close CMD window
- Don't close the CMD window while drive is mounted
- Press `Ctrl+C` in CMD to safely unmount

Mount as Windows Service (Auto-start)

```
# Install as service (runs automatically)
# Run CMD as Administrator, then:

nssm install rclone-personal "C:\rclone\rclone.exe" mount personal: X: --vfs-cache-mode wri

# Start service
nssm start rclone-personal

# Download NSSM from: https://nssm.cc/download
```

Real Use Case: Work with Files Locally

Mount your Drive, then use any program (Word, Excel, Photoshop) to directly open/edit files. Changes save automatically to Google Drive!

🤖 Automation & Scheduling

Create Batch Scripts

Daily Backup Script

Save as: `C:\Scripts\daily_backup.bat`

```
@echo off
echo Starting Daily Backup...
rclone copy C:\Users\Shantu\Documents personal:Backup/Documents -P
rclone copy C:\Users\Shantu\Pictures personal:Backup/Pictures -P
echo Backup Complete!
pause
```

Sync Multiple Drives Script

Save as: `C:\Scripts\sync_drives.bat`

```
@echo off
echo Syncing Personal to Work Backup...
rclone sync personal:Important work:Backup/Personal --dry-run
pause
echo Running actual sync...
rclone sync personal:Important work:Backup/Personal -P
echo Done!
pause
```

Schedule with Windows Task Scheduler

1. Press **Windows key**, type **Task Scheduler**
2. Click **"Create Basic Task"**
3. Name: `Daily Drive Backup`
4. Trigger: **Daily** at 11:00 PM
5. Action: **Start a program**
6. Program: `C:\Scripts\daily_backup.bat`
7. Click **Finish**

> ### Advanced: Backup on Network Connection
>
> 1. In Task Scheduler, set trigger to **"On an event"**
> 2. Log: `Microsoft-Windows-NetworkProfile/Operational`
> 3. Event ID: `10000` (Network Connected)
> 4. This backs up automatically when you connect to WiFi!

Sync on File Change (Real-time)

Download and install: SyncTrayzor (GUI for Syncthing)
Or use rclone's built-in bidirectional sync:

```
# Bidirectional sync (keeps both sides updated)
rclone bisync personal:Folder work:Folder --resync
```

```
# After first sync, run regularly:
rclone bisync personal:Folder work:Folder
```

📁 File Management

Delete Files

```
# Delete specific folder
rclone delete personal:OldFolder

# Delete files matching pattern
rclone delete personal:Downloads --include "*.tmp"

# Delete files older than 30 days
rclone delete personal:TempFiles --min-age 30d

# Delete with confirmation (safer)
rclone delete personal:Files --dry-run
# Then run without --dry-run if it looks correct
```

Empty Trash

```
# Empty trash for a remote
rclone cleanup personal:

# This permanently deletes trashed files!
```

Create Folders

```
# Create new folder
rclone mkdir personal:NewFolder

# Create nested folders
rclone mkdir personal:Projects/2024/January
```

Rename/Move Folders

```
# Rename folder
rclone moveto personal:OldName personal:NewName

# Move folder to different location
rclone moveto personal:Folder1 personal:Archive/Folder1
```

Check File Integrity

```
# Check if files match between two locations
rclone check personal:Files work:Backup/Files

# Show only differences
rclone check personal:Files work:Backup/Files --one-way

# Generate hash list (for verification)
rclone hashsum MD5 personal:ImportantFiles > hashes.txt
```

Size Analysis

```
# Get size of folder
rclone size personal:Projects

# Get size by file type
rclone size personal:Files --include "*.pdf"

# Get total space used
rclone about personal:
```

🔧 Troubleshooting

Common Issues & Solutions

Issue: "rclone is not recognized"

Solution:

- Check if `C:\rclone` is in PATH (see Installation section)
- Close and reopen Command Prompt
- Run: `echo %PATH%` to verify

Issue: Transfer Very Slow

Solutions:

```
# Increase transfers
rclone copy source: dest: --transfers 32

# Use fast-list for many files
rclone copy source: dest: --fast-list

# Increase chunk size for large files
rclone copy source: dest: --drive-chunk-size 64M
```

Issue: "Rate Limit Exceeded"

Solution:

```
# Add delay between API calls
rclone copy source: dest: --drive-pacer-min-sleep 100ms

# Reduce number of parallel transfers
rclone copy source: dest: --transfers 4
```

Issue: Mount Not Working

Solutions:

1. Install WinFsp first: [Download Here]
2. Run CMD as Administrator
3. Make sure drive letter isn't already in use
4. Try different cache mode: `--vfs-cache-mode full`

Issue: Authentication Expired

```
# Reconnect account
rclone config reconnect remote_name:
```

Issue: Duplicates Being Created

Solutions:

```
# Use --update to skip existing files
rclone copy source: dest: --update

# Or use sync instead of copy
rclone sync source: dest: --dry-run

# Remove duplicates after
rclone dedupe dest: --dedupe-mode newest
```

Enable Debug Logging

```
# Run with verbose logging
rclone copy source: dest: -vv --log-file=debug.txt

# Check debug.txt for detailed error messages
```

Test Connection

```
# Test if remote is working
rclone lsd personal:

# Check quota/space
rclone about personal:

# List configured remotes
rclone listremotes
```

📚 Complete Command Reference

Essential Commands

Command	Description	Example
`listremotes`	List all configured remotes	`rclone listremotes`
`lsd`	List directories only	`rclone lsd personal:`
`ls`	List all files with sizes	`rclone ls personal:Folder`
`lsl`	List with details (size, date)	`rclone lsl personal:Folder`
`lsf`	List in parseable format	`rclone lsf personal: --include "*.pdf"`
`copy`	Copy files (source stays)	`rclone copy source: dest:`
`sync`	Make dest identical to source	`rclone sync source: dest:`
`move`	Move files (deletes from source)	`rclone move source: dest:`
`delete`	Delete files	`rclone delete remote:Folder`
`mkdir`	Create directory	`rclone mkdir remote:NewFolder`
`rmdir`	Remove empty directory	`rclone rmdir remote:Folder`
`purge`	Delete folder and contents	`rclone purge remote:Folder`
`size`	Get size of folder	`rclone size remote:Folder`
`about`	Get quota info	`rclone about remote:`
`check`	Check if files match	`rclone check source: dest:`

`mount`	Mount as local drive	`rclone mount remote: X:`
`dedupe`	Remove duplicate files	`rclone dedupe remote:`
`cleanup`	Empty trash	`rclone cleanup remote:`

Important Flags

Flag	Purpose	Example
`-P`	Show progress	`rclone copy source: dest: -P`
`--dry-run`	Test without doing anything	`rclone sync source: dest: --dry-run`
`-v`	Verbose output	`rclone copy source: dest: -v`
`-vv`	Very verbose (debug)	`rclone copy source: dest: -vv`
`--include`	Include only matching files	`--include "*.pdf"`
`--exclude`	Exclude matching files	`--exclude "*.tmp"`
`--min-size`	Minimum file size	`--min-size 10M`
`--max-size`	Maximum file size	`--max-size 1G`
`--max-age`	Modified in last X time	`--max-age 7d`
`--min-age`	Not modified in last X time	`--min-age 30d`
`--update`	Skip newer files in dest	`rclone copy source: dest: --update`
`--bwlimit`	Limit bandwidth	`--bwlimit 1M`
`--transfers`	Parallel transfers	`--transfers 32`
`--log-file`	Save log to file	`--log-file=log.txt`

Quick Reference Card

📋 Common Tasks

```
# Backup folder
rclone copy C:\Docs personal:Backup -P

# Sync two drives
rclone sync personal: work:Mirror -P

# List all files
rclone ls personal:

# Check space
rclone about personal:

# Delete old files
rclone delete personal: --min-age 90d
```

⚡ Speed Tips

```
# Fast transfer
--transfers 32 --checkers 32

# Large files
--drive-chunk-size 64M

# Many small files
--fast-list

# Show progress
-P --stats 5s
```

🎯 **Pro Tips:**

Always use `--dry-run` before sync/delete operations | Use `-P` to monitor progress | Keep separate backups | Test with small folders first

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